The following listing shows a simple one-line change for adding the Vary header to the sample.

Listing 6.4 Listing 6.4 Setting the Vary response header

```
  if (options.allowOrigin) {
    var origin = req.headers['origin'];
    if (options.allowOrigin(origin)) {
      res.set('Access-Control-Allow-Origin', origin);
    }
    res.set('Vary', 'Origin');
  } else {
    res.set('Access-Control-Allow-Origin', '*');
  }
};
```

Setting Vary: Origin header

The Vary header is only necessary when the Origin header causes changes in the response. You don't need it when you return `Access-Control-Allow-Origin: *`, because the `*` value is the same regardless of what the value of the Origin header is. The Vary header should also be present even if the Access-Control-Allow-Origin header isn't in the response. If a request's origin is http://example.com, the response will not have an Access-Control-Allow-Origin header, because http://example.com isn't in the list of whitelisted headers. But the response should still contain a Vary header, because the response can change, based on the Origin header value.

6.3.4 Null origin

So far we've only discussed origins in terms of URLs with a scheme, host, and port. But when configuring the acceptable origin values, it's important to keep in mind that `null` is also a valid origin value. A `null` origin value typically indicates that the request is coming from a file on a user's computer, rather than from a website. It can also mean the request came from a redirect (which we'll cover in more detail later).

Making requests from a file on your computer (which I'll refer to as a *local file*) is useful if you want to test CORS requests without uploading files to a live server. It can be a lot easier to test locally than to upload new code every time you'd like to test a request. Testing code is an iterative process: the developer writes code, tests the results, and writes more code based on the results. When working with a local file, a developer can iterate very quickly: trying new code is as easy as saving a file and refreshing the web browser.

In contrast, if CORS requests required an actual origin for every request, the developer velocity would be much slower. Each change would need to be uploaded to a web server, which can be slow (at least when compared with saving a file locally). The developer would also need a separate test server to test changes on, because uploading to the live server could break other users. Even if you don't anticipate using this technique to test your own API, opening up your API to `null` origins can make it easier for your users to test their code against your API.